3264.TWO - Ardentec Corporation | Full Deep Dive

Full investment research | Model assumptions and source evidence

October 11, 2026 selective-pricing update | Original September 21 model retained; current Framework 8.0 Rule 16b applied to new evidence | NOT HELD | STRUCTURAL - AI ASIC and optical testing; TEMPORAL - utilization recovery | High Conviction

Analytical Base: 47/60 (78%). Valuation: 20x forward four-quarter IFRS-style operating profit with depreciation-transition reserve.

Snapshot: TWD 268.50 per share on September 18, 2026 TPEx close; 473.920 million shares, implying TWD 127,247.52 million equity value. Source [ARD-8]. Price is a dated header snapshot, not a valuation input in the body.

Active Q4 2027 ceiling: TWD 416.180997/share, versus TWD 413.521432 before this evidence (+0.6432%). Q1 2028: TWD 437.437456/share. Excel Base 78 and analytical Base 78 are unchanged.

All financial model amounts are TWD millions, shares in millions, unless stated. Calendar quarters. User alias 3264.TW is TPEx-listed 3264.TWO. Reported amounts in TWD thousands are converted to millions. Header shares exclude subsidiary-held treasury shares.

October 11 update: selective test pricing, after costs and overlap

The visible lead of the October 12, 02:26 Taipei report (October 11, 14:26 Toronto) explicitly names Ardentec (欣銓) among beneficiaries of tight packaging/testing capacity. It attributes the pressure to longer AI test times and customer-designed chips, reporting industry price increases of 10-30% and much larger charges for some small rush orders. Only the lead is accessible. This is an attributed industry check, not Ardentec guidance or proof of a company-wide increase. The small-customer doubling anecdote is excluded. [ARD-10]

ASTRA assigns **55% probability** to a durable **20% conditional premium**, bounded at **10-30%**, on **5% of total baseline test-service revenue** that can be newly contracted or repriced. The existing established, ASIC and optical vectors already contain growth, utilization and favorable margin assumptions. **50% overlap** is removed from the probability-weighted premium to avoid valuing the same economics twice. The source does not identify Ardentec's eligible contracts or realized rates; probability, cohort, overlap, timing and costs are explicit model assumptions.

Recognition phases at **25% / 50% / 75% / 100% in Q1-Q4 2027**, remains **100% in 2028**, fades to **50% in 2029**, and reaches **zero in 2030**. These phases reflect billing/acceptance on new or repriceable work, not a claim that the article gives that schedule. Fixed-price backlog does not automatically reset. The fade allows for competitive capacity and customer bargaining responses. There is no new physical-output, equipment or TAM increment.

Incremental revenue = baseline test revenue x 5% eligible x 20% premium x 55% probability x (1 - 50% overlap) x phase. Manufacturing costs consume **20%** of this net incremental revenue and support costs consume **5%**, leaving **75% incremental OP capture**. This is a marginal price/cost calculation, not the company's operating margin. Baseline R&D and SG&A remain tied to baseline revenue; only the specified support expense is added, so the old cost ratios do not automatically charge a second expense layer on the premium. Baseline factory depreciation, the separate depreciation-transition reserve and dilution schedule remain intact.

At full phase, expected incremental revenue is **0.275% of baseline test sales**, with an OP increment of **0.20625% of baseline sales**. The 10-30% conditional range translates into **0.1375-0.4125%** expected revenue uplift under the same remaining assumptions. Longer qualified-test queues can support pricing because waiting delays customers' chip shipments, but customer alternatives, qualification, contract duration and usable test throughput govern capture.

The active Q4 2027 window gains **TWD 63.530362M** of F4Q operating profit. At the retained **20x** and **477.750M diluted shares**, the ceiling moves from **TWD 413.521432** to **TWD 416.180997**, or **+0.6432%**. The 10% and 30% premium bounds give **TWD 414.851214-417.510779**. These are sensitivities at the same probability and cohort, not extra weighted cases. If the new pricing branch fails, the model returns to the prior ceiling path.

The important gap is Ardentec-specific contract realization. Monitor test-service prices and collections, cohort share, acceptance dates and incremental labor/maintenance/consumable costs. Reduce or defer the branch if observed margin gains merely reproduce the baseline utilization/mix case, or if customers move work to competing capacity. This uncertainty is retained alongside the active bounded increment.

1. Thesis, bottleneck and expanded TAM

Ardentec earns testing-service revenue from semiconductor manufacturers and chip designers. AI ASIC complexity, longer test requirements and optical-electrical integration can raise test content per chip. The immediate economic driver is more high-end equipment hours sold at profitable utilization.

Two facilities matter for different reasons: Longtan supports AI ASIC wafer testing; wholly owned Giga Solution's Hukou facility supports silicon-photonics-related products. The model separates those ramps from the established testing business and does not count all new capacity as instantly full. [ARD-2, ARD-5]

The user's LIWEI post highlights the optical opportunity and proposes 40% gross margin for high-end ASIC testing. The latest reported consolidated gross margin is already about40.2%; the post's product-level assumption is not formal guidance or proof of a fivefold investment return. [ARD-1, ARD-9]

Competition includes integrated assembly/test providers, independent test specialists and customers' own testing operations. Service quality, qualification, uptime and capital efficiency decide outsourced share. Giga's publicly described partnership with Fabrinet and Raytek supports an integrated offering, but does not guarantee exclusive customer allocation or ownership of the partners' sales. [ARD-6]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
Phase 1 - current, 100% relevance	Established wafer sort, final test and mixed-signal/RF work	Actual revenue and equipment utilization
Phase 2 - 2027-2028, 50% conceptual weight	AI ASIC wafer-test outsourcing and optical qualification	Longtan and Hukou throughput ramps
Phase 3 - 2029-2030, 20% conceptual weight	More EIC/PIC and Micro LED testing intensity	Additional equipment and qualified programs required

The relevant market is outsourced testing spend, not total semiconductor sales or total optical-module sales. Higher Goldman module forecasts can increase test activity, but no disclosed fee per module or share of customer programs lets us translate that TAM directly. Use equipment-years times utilization times service revenue per utilized machine-year as a Plan B optical proxy. Do not apply module ASPs to Ardentec test services.

2. Latest evidence and financial anchor

Q2 exact revenue was4,533.667M, GP1,820.969M and OP1,377.355M. H1 totals were8,517.069M,3,348.888M and2,473.880M respectively. These are operating figures, separate from investment disposal, associate income and FX below the operating line. [ARD-1]

May’s company presentation showed Q1 revenue3,983M, GP1,528M and OP1,097M. Local reporting of the May call described Longtan’s Q3 start and optical revenue still small ahead of a larger2027 contribution. July materials then documented completed customer validation and July production at Longtan. [ARD-2, ARD-3, ARD-7]

The July deck lists more than150 EIC/PIC test systems, more than50 Micro LED systems under construction, and Giga’s new Hukou production site. [ARD-2] These are different equipment/program categories, not200 interchangeable fully utilized systems.

Issuer monthly sales were1,766.652M in July and1,794.449M in August. The modeled Q3 total5,350M implies September1,788.899M and17.99% sequential growth. This is consistent with a double-digit ramp without assuming another jump in every remaining month. [ARD-4]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026	IFRS, H1 less Q2	3,983.40	1,527.92	1,096.53	ARD-1
Q2 2026	Reported IFRS	4,533.67	1,820.97	1,377.36	ARD-1

Q2 GP1,820.969 less R&D156.189 and selling/admin287.425 equals OP1,377.355. H1 product mix was70.3% wafer sort,29.0% final test and0.7% other. [ARD-1, ARD-2] The new revenue vectors below are analytic growth buckets, not company-reported segments; legacy revenue excludes the separately modeled incremental Longtan/optical ramp. Gross margins include normal factory depreciation.

July local-language call reporting described sequential H2 growth, rising utilization and the possibility of Hukou optical capacity approaching full use in2027. [ARD-5] That is a conditional operating outlook, not an exact sales or margin guide. Q3/Q4 revenue5,350/5,850M gives FY2026 sales19,717.069M; all later quarters are analyst assumptions. There is no assumption that each additional floor has identical installed equipment or revenue.

Balance sheet and capital allocation

June cash was1,825.775M; current amortized-cost financial assets were about1,489M. Bank debt was13,477.533M including the current portion of long debt, giving net bank debt about11,651.758M before deposits and leases. H1 operating cash flow2,541M did not cover7,007M equipment/property investment: free cash flow was approximately minus4,466M. [ARD-1, ARD-2] Expansion therefore requires capital discipline, not just accounting profit. Raytek is a24.99% equity-method investment; its entire revenue and OP are excluded from consolidated vectors. [ARD-1]

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	16	Qualified wafer and optical test service platforms with rising complexity
Cycle Stage	9	Reported revenue, OP and utilization already improving
TAM Expansion	9	AI ASIC outsourcing plus EIC/PIC and Micro LED testing
Catalysts	5	Longtan production and Hukou optical ramp
Macro	9	AI infrastructure and optical content support testing demand
Alpha	4	Test-hour and cash-return bridge separates economics from module TAM
Risks	-5	Loading, capital/debt, depreciation, customer mix, technology competition
Total	47/60 = 78%	High Conviction

4. Quarterly operating model

Established test services, incremental AI ASIC wafer-test programs and the optical expansion form nonoverlapping vectors. R&D starts at3.5% and SG&A6.2%, close to Q2 functional costs. Gross margins assume ordinary depreciation already. A separate expense reserve accounts for the fading benefit of the disclosed historical equipment-life extension; it is not another full depreciation charge.

Revenue vector	Driver and source	Confidence
Established test services	ARD-1 actual base; ARD-4 monthly sales	MEDIUM near-term; LOW long-term
Incremental high-end AI ASIC wafer test	ARD-2 Longtan start; qualified service expansion	MEDIUM ramp direction; LOW exact revenue allocation
Optical / EIC-PIC expansion	ARD-2 system count; ARD-5 loading direction	LOW fee, utilization and new-equipment proxies

Established services grow from the H2 annualized17,400M base to18,400M in2027, then about5% yearly. Incremental ASIC testing reaches6,500M in2027, about35% above Q4’s annualized4,800M ramp, then9,500/12,500/15,500M as more qualified equipment comes online. Optical revenue is1,800/3,500/5,500/7,500M for2027-2030. For2027,150 systems x70% utilized-equipment-years x17.143M revenue per utilized system-year approximates1,800M. Later planning cases require250/350/450 system equivalents at78%/82%/85% utilization and17.95/19.16/19.61M service revenue per utilized year. Only the greater-than150 count is disclosed; later counts, fees and utilization are analyst proxies. They already allow for ramp uncertainty; no second confidence discount is applied. Physical test time, parallelism and product complexity make system equivalents imperfect.

Baseline physical and commercial drivers

The following physical revenue baseline is unchanged. The separate selective-pricing branch below adds to consolidated sales without adding output.

Year	Established	AI ASIC incremental	Optical expansion	Baseline total
2027	18,400	6,500	1,800	26,700
2028	19,320	9,500	3,500	32,320
2029	20,286	12,500	5,500	38,286
2030	21,300	15,500	7,500	44,300

Vector revenue, gross margin and gross profit

These three vector tables remain the baseline before selective pricing. For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

Established test services

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	4,300.00	40.50%	1,741.50
Q4 2026	4,400.00	40.50%	1,782.00
Q1 2027	4,232.00	41.00%	1,735.12
Q2 2027	4,508.00	41.00%	1,848.28
Q3 2027	4,692.00	41.00%	1,923.72
Q4 2027	4,968.00	41.00%	2,036.88
Q1 2028	4,443.60	41.50%	1,844.09
Q2 2028	4,733.40	41.50%	1,964.36
Q3 2028	4,926.60	41.50%	2,044.54
Q4 2028	5,216.40	41.50%	2,164.81
Q1 2029	4,665.78	42.00%	1,959.63
Q2 2029	4,970.07	42.00%	2,087.43
Q3 2029	5,172.93	42.00%	2,172.63
Q4 2029	5,477.22	42.00%	2,300.43
Q1 2030	4,899.00	42.00%	2,057.58
Q2 2030	5,218.50	42.00%	2,191.77
Q3 2030	5,431.50	42.00%	2,281.23
Q4 2030	5,751.00	42.00%	2,415.42

Incremental high-end AI ASIC wafer test

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	900.00	48.00%	432.00
Q4 2026	1,200.00	48.00%	576.00
Q1 2027	1,495.00	48.00%	717.60
Q2 2027	1,592.50	48.00%	764.40
Q3 2027	1,657.50	48.00%	795.60
Q4 2027	1,755.00	48.00%	842.40
Q1 2028	2,185.00	49.00%	1,070.65
Q2 2028	2,327.50	49.00%	1,140.47
Q3 2028	2,422.50	49.00%	1,187.03
Q4 2028	2,565.00	49.00%	1,256.85
Q1 2029	2,875.00	49.00%	1,408.75
Q2 2029	3,062.50	49.00%	1,500.62
Q3 2029	3,187.50	49.00%	1,561.88
Q4 2029	3,375.00	49.00%	1,653.75
Q1 2030	3,565.00	49.00%	1,746.85
Q2 2030	3,797.50	49.00%	1,860.77
Q3 2030	3,952.50	49.00%	1,936.72
Q4 2030	4,185.00	49.00%	2,050.65

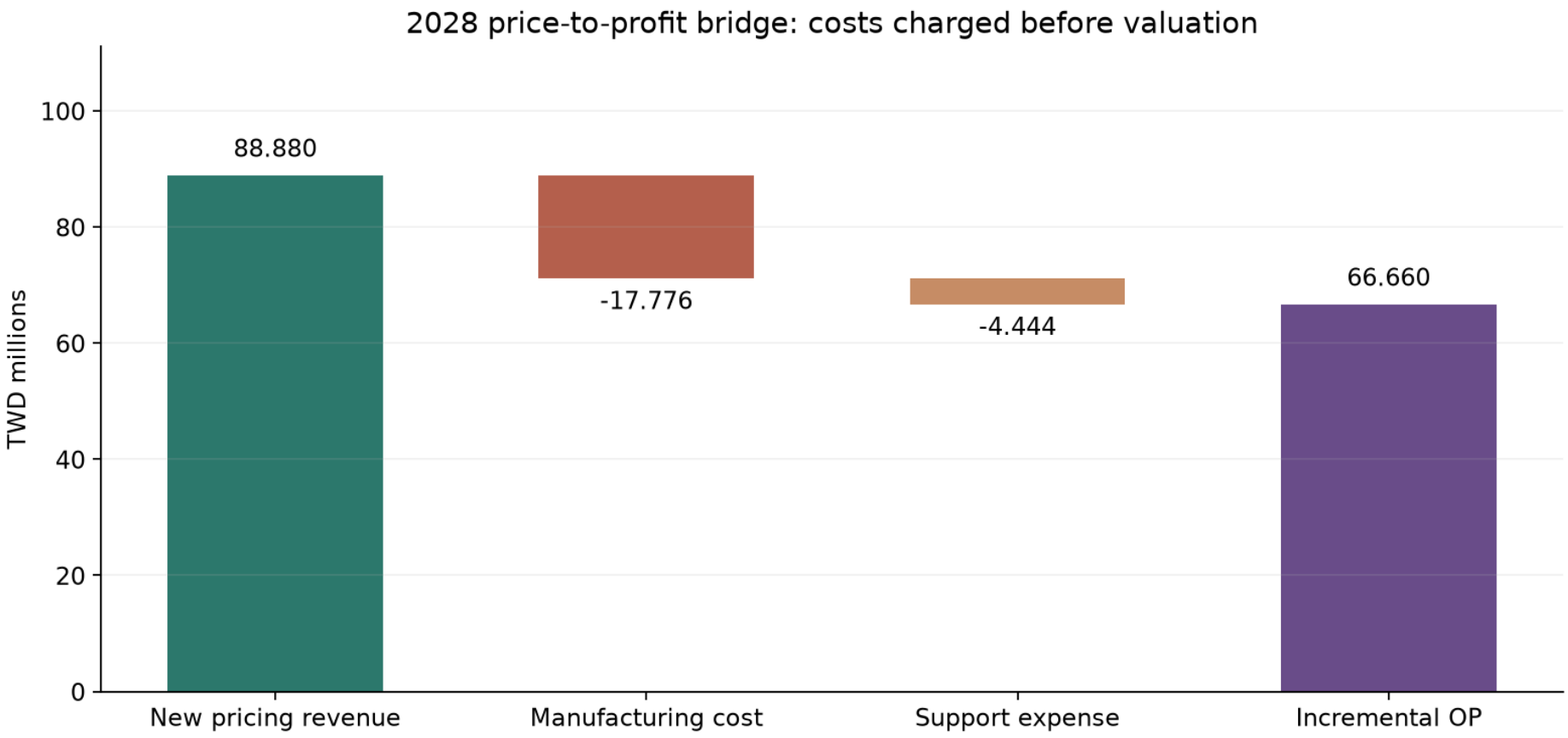
Optical / EIC-PIC expansion

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	150.00	25.00%	37.50
Q4 2026	250.00	30.00%	75.00
Q1 2027	414.00	40.00%	165.60
Q2 2027	441.00	40.00%	176.40
Q3 2027	459.00	40.00%	183.60
Q4 2027	486.00	40.00%	194.40
Q1 2028	805.00	44.00%	354.20
Q2 2028	857.50	44.00%	377.30
Q3 2028	892.50	44.00%	392.70
Q4 2028	945.00	44.00%	415.80
Q1 2029	1,265.00	46.00%	581.90
Q2 2029	1,347.50	46.00%	619.85
Q3 2029	1,402.50	46.00%	645.15
Q4 2029	1,485.00	46.00%	683.10
Q1 2030	1,725.00	47.00%	810.75
Q2 2030	1,837.50	47.00%	863.62
Q3 2030	1,912.50	47.00%	898.88
Q4 2030	2,025.00	47.00%	951.75

Selective pricing added to the test-service baseline

Values are TWD millions. Incremental revenue already includes the 55% probability and 50% overlap deduction. A 5% eligible cohort is applied to the sum of the three existing vectors once.

Quarter	Phase	Baseline sales	New pricing revenue	Incremental COGS	Incremental GP	Support cost	Incremental OP
Q3 2026	0%	5,350.000	0.000	0.000	0.000	0.000	0.000
Q4 2026	0%	5,850.000	0.000	0.000	0.000	0.000	0.000
Q1 2027	25%	6,141.000	4.222	0.844	3.378	0.211	3.166
Q2 2027	50%	6,541.500	8.995	1.799	7.196	0.450	6.746
Q3 2027	75%	6,808.500	14.043	2.809	11.234	0.702	10.532
Q4 2027	100%	7,209.000	19.825	3.965	15.860	0.991	14.869
Q1 2028	100%	7,433.600	20.442	4.088	16.354	1.022	15.332
Q2 2028	100%	7,918.400	21.776	4.355	17.420	1.089	16.332
Q3 2028	100%	8,241.600	22.664	4.533	18.132	1.133	16.998
Q4 2028	100%	8,726.400	23.998	4.800	19.198	1.200	17.998
Q1 2029	50%	8,805.780	12.108	2.422	9.686	0.605	9.081
Q2 2029	50%	9,380.070	12.898	2.580	10.318	0.645	9.673
Q3 2029	50%	9,762.930	13.424	2.685	10.739	0.671	10.068
Q4 2029	50%	10,337.220	14.214	2.843	11.371	0.711	10.660
Q1 2030	0%	10,189.000	0.000	0.000	0.000	0.000	0.000
Q2 2030	0%	10,853.500	0.000	0.000	0.000	0.000	0.000
Q3 2030	0%	11,296.500	0.000	0.000	0.000	0.000	0.000
Q4 2030	0%	11,961.000	0.000	0.000	0.000	0.000	0.000



Ardentec 2028 modeled selective-pricing revenue, manufacturing cost, support expense and operating-profit bridge

Calculated from the explicit ASTRA pricing assumptions; not source-reported company guidance.

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A incl. support	Other opex	OP	OP margin
Q3 2026	Model	5,350.000	2,211.000	187.250	331.700	0.000	1,692.050	31.627%
Q4 2026	Model	5,850.000	2,433.000	204.750	362.700	0.000	1,865.550	31.890%
Q1 2027	Model	6,145.222	2,621.698	214.935	368.671	76.122	1,961.969	31.927%
Q2 2027	Model	6,550.495	2,796.276	228.953	392.940	76.122	2,098.261	32.032%
Q3 2027	Model	6,822.543	2,914.154	238.298	409.212	76.122	2,190.522	32.107%
Q4 2027	Model	7,228.825	3,089.540	252.315	433.531	76.122	2,327.572	32.198%
Q1 2028	Model	7,454.042	3,285.298	260.176	432.171	205.283	2,387.668	32.032%
Q2 2028	Model	7,940.176	3,499.556	277.144	460.356	205.283	2,556.773	32.200%
Q3 2028	Model	8,264.264	3,642.396	288.456	479.146	205.283	2,669.510	32.302%
Q4 2028	Model	8,750.398	3,856.654	305.424	507.331	205.283	2,838.616	32.440%
Q1 2029	Model	8,817.888	3,959.964	308.202	502.535	317.828	2,831.398	32.110%
Q2 2029	Model	9,392.968	4,218.222	328.302	535.309	317.828	3,036.783	32.330%
Q3 2029	Model	9,776.354	4,390.395	341.703	557.158	317.828	3,173.706	32.463%
Q4 2029	Model	10,351.434	4,648.653	361.803	589.932	317.828	3,379.090	32.644%
Q1 2030	Model	10,189.000	4,615.180	356.615	580.773	317.828	3,359.964	32.976%
Q2 2030	Model	10,853.500	4,916.170	379.873	618.649	317.828	3,599.820	33.167%
Q3 2030	Model	11,296.500	5,116.830	395.378	643.901	317.828	3,759.723	33.282%
Q4 2030	Model	11,961.000	5,417.820	418.635	681.777	317.828	3,999.579	33.439%

Annual trajectory cross-check

The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin
2027	26,747.084	8,578.325	32.072%
2028	32,408.880	10,452.568	32.252%
2029	38,338.643	12,420.976	32.398%
2030	44,300.000	14,719.086	33.226%

Accounting cross-check: the company extended certain machine lives from7 to9 years effective2025. The interim note estimates depreciation reductions of771.314M in2026 and466.826M in2027, then49.818M additional expense in2028, with2,086.421M additional expense in later years collectively. [ARD-1] Keeping gross margins on a2026 cost basis requires incremental reserves of304.488M in2027 and821.132M in2028. For2029/2030, an assumed500M annual reversal yields1,271.314M annual reserve versus that basis; the exact later timing is not guided. Without those reserves, ceilings would overstate the continuing benefit. Normal depreciation on new equipment is already embedded in vector margins. Separately, if AI ASIC GM is only40% instead of48% in2027, OP falls520M and the Q1 2027 ceiling falls about21.79TWD/share, holding all other inputs fixed.

5. Forward four-quarter ceiling and derivations

20x forward-four-quarter IFRS-style operating profit with depreciation-transition reserve is the portfolio owner-selected multiple as of September 21, 2026, held constant across the model. This is the framework operating-profit convention, not an observed market P/E. A one-turn change changes each ceiling by 5%. The depreciation-transition reserve remains intact. Equity-method Raytek upside is not separately added; no net-debt deduction or after-tax conversion enters the framework ceiling.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	7,617.830875	20x	477.000	319.405907
Q4 2026	Q4 2026 to Q3 2027	8,116.303273	20x	477.150	340.199236
Q1 2027	Q1 2027 to Q4 2027	8,578.324836	20x	477.300	359.452120
Q2 2027	Q2 2027 to Q1 2028	9,004.023383	20x	477.450	377.171364
Q3 2027	Q3 2027 to Q2 2028	9,462.535461	20x	477.600	396.253579
Q4 2027	Q4 2027 to Q3 2028	9,941.523562	20x	477.750	416.180997
Q1 2028	Q1 2028 to Q4 2028	10,452.568000	20x	477.900	437.437456
Q2 2028	Q2 2028 to Q1 2029	10,896.298301	20x	478.050	455.864378

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2028	Q3 2028 to Q2 2029	11,376.307458	20x	478.200	475.797050
Q4 2028	Q4 2028 to Q3 2029	11,880.502519	20x	478.350	496.728442
Q1 2029	Q1 2029 to Q4 2029	12,420.976437	20x	478.500	519.163069
Q2 2029	Q2 2029 to Q1 2030	12,949.541637	20x	478.650	541.086039
Q3 2029	Q3 2029 to Q2 2030	13,512.578480	20x	478.800	564.435191
Q4 2029	Q4 2029 to Q3 2030	14,098.596418	20x	478.950	588.729363
Q1 2030	Q1 2030 to Q4 2030	14,719.086000	20x	479.100	614.447339

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: (1,692.050000 + 1,865.550000 + 1,961.969453 + 2,098.261422) x 20 / 477.000 = **TWD 319.405907 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q4 2026: OP window Q4 2026 to Q3 2027: (1,865.550000 + 1,961.969453 + 2,098.261422 + 2,190.522398) x 20 / 477.150 = **TWD 340.199236 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q1 2027: OP window Q1 2027 to Q4 2027: (1,961.969453 + 2,098.261422 + 2,190.522398 + 2,327.571563) x 20 / 477.300 = **TWD 359.452120 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q2 2027: OP window Q2 2027 to Q1 2028: (2,098.261422 + 2,190.522398 + 2,327.571563 + 2,387.668000) x 20 / 477.450 = **TWD 377.171364 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q3 2027: OP window Q3 2027 to Q2 2028: (2,190.522398 + 2,327.571563 + 2,387.668000 + 2,556.773500) x 20 / 477.600 = **TWD 396.253579 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q4 2027: OP window Q4 2027 to Q3 2028: (2,327.571563 + 2,387.668000 + 2,556.773500 + 2,669.510500) x 20 / 477.750 = **TWD 416.180997 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q1 2028: OP window Q1 2028 to Q4 2028: (2,387.668000 + 2,556.773500 + 2,669.510500 + 2,838.616000) x 20 / 477.900 = **TWD 437.437456 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q2 2028: OP window Q2 2028 to Q1 2029: (2,556.773500 + 2,669.510500 + 2,838.616000 + 2,831.398301) x 20 / 478.050 = **TWD 455.864378 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q3 2028: OP window Q3 2028 to Q2 2029: (2,669.510500 + 2,838.616000 + 2,831.398301 + 3,036.782657) x 20 / 478.200 = **TWD 475.797050 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q4 2028: OP window Q4 2028 to Q3 2029: (2,838.616000 + 2,831.398301 + 3,036.782657 + 3,173.705562) x 20 / 478.350 = **TWD 496.728442 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

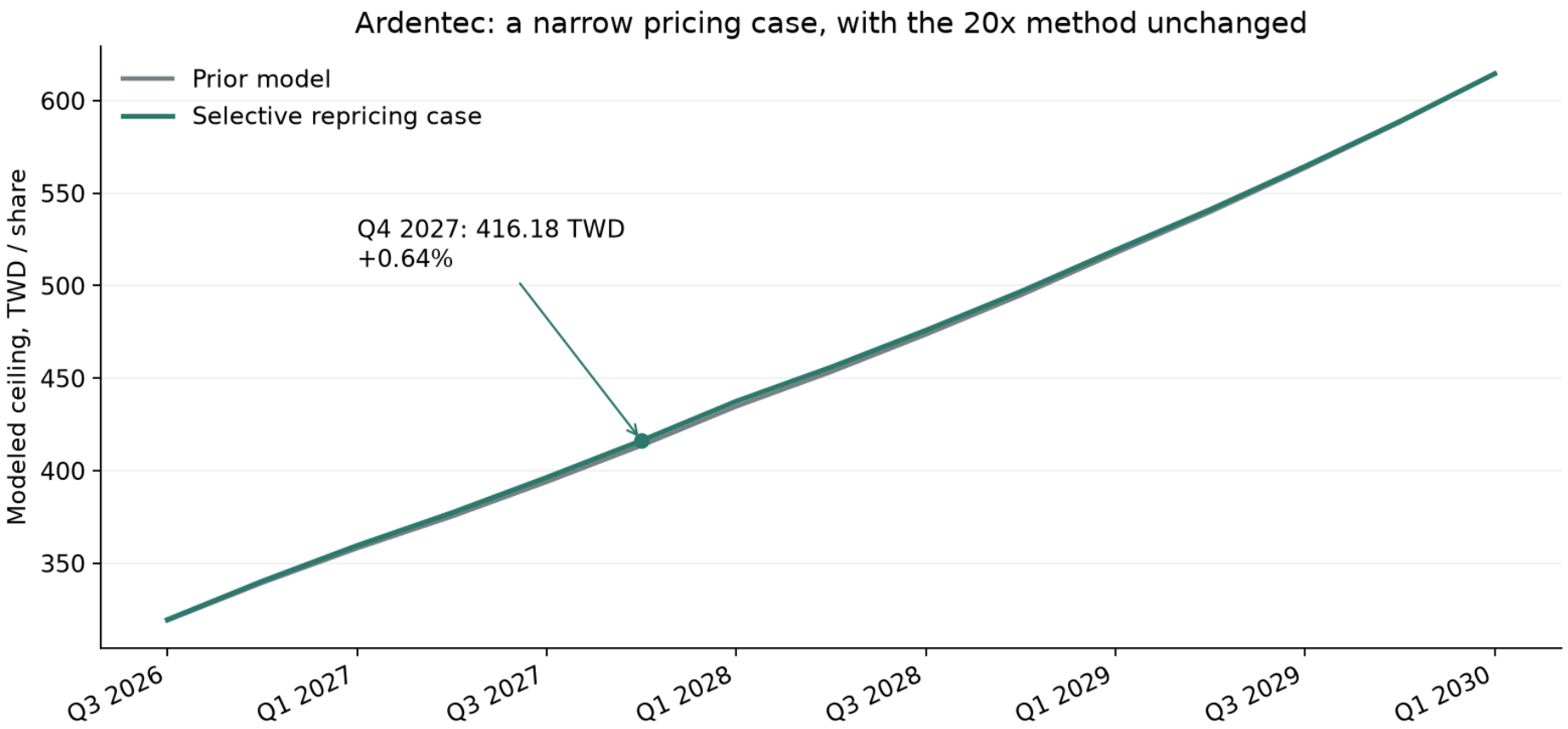
Q1 2029: OP window Q1 2029 to Q4 2029: (2,831.398301 + 3,036.782657 + 3,173.705562 + 3,379.089918) x 20 / 478.500 = **TWD 519.163069 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q2 2029: OP window Q2 2029 to Q1 2030: (3,036.782657 + 3,173.705562 + 3,379.089918 + 3,359.963500) x 20 / 478.650 = **TWD 541.086039 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q3 2029: OP window Q3 2029 to Q2 2030: (3,173.705562 + 3,379.089918 + 3,359.963500 + 3,599.819500) x 20 / 478.800 = **TWD 564.435191 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q4 2029: OP window Q4 2029 to Q3 2030: (3,379.089918 + 3,359.963500 + 3,599.819500 + 3,759.723500) x 20 / 478.950 = **TWD 588.729363 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.

Q1 2030: OP window Q1 2030 to Q4 2030: (3,359.963500 + 3,599.819500 + 3,759.723500 + 3,999.579500) x 20 / 479.100 = **TWD 614.447339 per share**. Each OP includes baseline vector GP plus incremental pricing GP, less baseline R&D/SG&A, incremental support and the retained depreciation-transition reserve.



Ardentec exact quarterly ceiling path before and after the narrow selective-pricing branch

The small separation reflects the limited eligible cohort and overlap allowance. This chart shows model calculations with the same shares and valuation method.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

Q2 diluted weighted-average shares were475.437M and H1 diluted shares476.617M; outstanding common shares excluding subsidiary-held stock were473.920M, versus490.1701M issued. Model shares begin477M and rise0.15M each quarter for employee and potential issuance dilution. Do not use issued shares and then add treasury shares again. [ARD-1]

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	437.437456
5% more shares	416.607101
10% more shares	397.670414

6. Cross-book relationships and monitoring

CHPT sells test interfaces and Ardentec sells testing capacity. Their growth may correlate, but neither receives the whole AI chip market or optical-module selling price.

The NAND/eSSD expansion can increase storage-controller testing, although wafer availability and test outsourcing determine orders. Rising NAND ASPs are not applied directly to service fees.

Giga's Micro LED test work connects conceptually to AUO's optical architecture work. There is no disclosed direct AUO customer relationship assumed in this model.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Loading and qualification	-1	New sites remain underutilized or customer acceptance slips
Capital spending and debt	-1	Borrowing and replacement investment outrun operating cash flow
Depreciation economics	-1	Longer accounting lives overstate durable returns; reversal accelerates
Customer and mix	-1	Large customers retain negotiating power or shift test work internally
Technology and competition	-1	Competing optical platforms or test methods limit content and share

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
October2026 Q3 report	Revenue near5,350M and evidence of profitable new-site loading	Reconcile apparent GP gains with depreciation and one-time effects
2027 new/repriceable test contracts	Realized price capture and collections beyond baseline mix/utilization	Reduce or defer the pricing branch if the narrow eligible cohort or economics fail
2027 Hukou/Longtan ramp	Qualified hours and cash collections support the new vectors	Cut optical revenue if equipment is installed but not productive
2028 depreciation and investment	New sales cover the fading useful-life benefit and capex	Raise reserve or lower margin if economics weaken
2029-2030 capacity expansion	New system-equivalent capacity is supported by accepted programs	Avoid extrapolating the initial optical count indefinitely

May and July official presentations and full Q1/Q2 financial reports were downloaded and reviewed; local call reporting supplements the management packages. Full verbatim conference transcripts were not available in reviewed public sources. Incremental ASIC/optical splits, machine fees and future utilization are analytic estimates. No precise customer share or long-term contracted optical revenue is disclosed. The complete artifact retains these limitations beside its full quarterly model.

8. Sources and snapshot

- **ARD-1** - Q2 report2026. Chinese interim financial statements. Downloaded; income, balance sheet, shares, associate and useful-life notes. Q1 report1150331C also reviewed.
- **ARD-2** - July24,2026. Q2 Mandarin management presentation.pdf). Downloaded; financials, Longtan and optical capacity; charts visually reviewed.
- **ARD-3** - May4,2026. Q1 Mandarin management presentation.pdf). Downloaded; prior financial and operating cycle.
- **ARD-4** - September2026. Issuer monthly revenue. Reported January-August2026 revenue.
- **ARD-5** - July24,2026. Liberty Times management-call reporting. Facility timing and conditional2027 utilization outlook.
- **ARD-6** - AccessedSeptember21,2026. Giga Solution optical testing services. Primary confirmation of Fabrinet/Raytek service partnership.
- **ARD-7** - May4,2026. Economic Daily management-call reporting. Q1 discussion of new-site timing and optical revenue scale.
- **ARD-8** - September18,2026. TPEx closing price. Header snapshot only; cross-checked price history.
- **ARD-9** - September18,2026. User-supplied LIWEI optical-testing thesis. User post;40% ASIC margin is an assumption, not formal guidance.
- **ARD-10** - October 12, 2026 02:26 Taipei / October 11 14:26 Toronto. Economic Daily visible lead: test-capacity scarcity and selective price increases. Explicitly names Ardentec; attributed industry checks, not issuer guidance. Full text remains gated. BQYV relay post IDs 2109430357724631519, 2109430442915152261. No unseen text used.

Model snapshot: October 11, 2026 selective-pricing update. Original September 21 operating baseline retained. 3264.TWO. Framework 8.0 Rule 16b evidence treatment; existing valuation method preserved. Base 47/60 (78%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/3264TW_Ardentec_DeepDive.pdf and .md. The companion company worksheet uses this same model and links the ceiling path to the master portfolio.